8) Sigmoid function

```
import numpy as np
from sklearn.datasets import
load_breast_cancer
from sklearn.model_selection import train_test_split
#load Breast cancer dataset
data= load_breast_cancer()
X = data.data        # 30 features
Y = data.target      # 0 or 1
# Train-Test split
X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size =0.2, random_state =42)
# Min-Max Normalization (using TRAIN data)
X_min = X_train.min(axis=0)
X_max = X_train.max(axis=0)
X_train = (X_train -X_min) / (X_max -X_min)
X_test = (X_test -X_min) / (X_max - X_min)
# Initialize single neuron parameters
np.random.seed(1)
Weights = np.random.rand(30)
bias =0.0
Learning_rate= 0.01
epochs = 10
# Sigmoid Activation function
def sigmoid (x):
    return 1/(1+ np.exp(-x))
# Training phase
print ("TRAINING PHASE (SIGMOID FUNCTION)")
print ("=" * 60)
for epoch in range (1, epochs +1):
    Correct = 0
    for i in range (len (X_train)):
        net_input= np.dot (X_train [i] , weights) +bias
    output = sigmoid (net_input)
    prediction = 1 if output >= 0.5 else 0
    error = Y_train [i] -output
    #update
    Weights + = Learning_rate * error * X_train [i]
    bias + = learning_rate * error
    if prediction == Y_train [i]:
        Correct += 1
```

```
epoch_accuracy = (correct / len(X_train)) * 100
print(f"Epoch {epoch}")
print(f"Training Accuracy: {epoch_accuracy:.2f}%")
print(f"Bias: {bias:.4f}")
print(f"Weights (first 5 shown): {weights[:5]}")
print("-" * 60)
# Final Training Accuracy
print("\n FINAL TRAINING ACCURACY")
print("=" * 60)
Correct = 0
for i in range(len(X_train)):
    net_input = np.dot(X_train[i], weights) + bias
    prediction = 1 if sigmoid(net_input) >= 0.5 else 0
    if prediction == Y_train[i]:
        correct += 1
print(f"Final Training Accuracy: {(correct / len(X_train))
    * 100:.2f}%")
# Testing phase
print("\n TESTING PHASE")
print("=" * 60)
Correct = 0
for i in range(len(X_test)):
    net_input = np.dot(X_test[i], weights) + bias
    prediction = 1 if sigmoid(net_input) >= 0.5 else 0
    if prediction == Y_test[i]:
        correct += 1
print(f" Final Test Accuracy: {(correct / len(X_test)) * 100
    :.2f}%")
```

OUTPUT

TRAINING PHASE (SIGMOID FUNCTION)

Epoch 1

Training Accuracy : 59.34%

Bias : -0.4929

Weights (first 5 shown): [-0.02760619 0.40131814 -0.45083322 -0.04788781 -0.16563215]

Epoch 2

Traing Accuracy : 64.84%

Bias : -0.4929

Weights (first 5 shown): [-0.14602605 0.37306314 -0.57629913 -0.14191284 -0.12678329]

Epoch 3

Training Accuracy : 80.88%

Bias : 0.2451

Weights (first 5 shown): [-0.22533452 0.35668117 -0.66160236 -0.26136157 -0.07779486]

Epoch 4

Training Accuracy : 85.49%

Bias : 0.5618

Weights (first 5 shown): [-0.28924431 0.33693613 -0.73065544 -0.33449599 -0.03701232]

Epoch 5

Training Accuracy : 87.03%

Bias : 0.8405

Weights (first 5 shown): [-0.34237253 0.3147894 -0.7882509 -0.39604775 -0.00289971]

FINAL TRAINING ACCURACY

Final Training Accuracy : 88.57%

TESTING PHASE

Final Test Accuracy : 90.35%